

COL2010: Data Science

End-Semester Examination — Module 2: Data Cleaning and Preprocessing

SET F

Time: 3 Hours

Total Marks: 100

Instructions to candidates.

- This paper contains **15** questions. Attempt **all** questions.
- All numerical answers must retain at least four significant figures unless an exact closed form is requested.
- All claims of the form “prove”, “show”, or “derive” require a fully rigorous argument; a correct final formula with no justification earns no credit.
- State any assumption you invoke that is not already given in the question.
- Calculators are permitted; symbolic computer algebra devices are not.

Q1. For weights $w_1, \dots, w_d > 0$, define the weighted standardized Euclidean distance $d_w(x, y) = \sqrt{\sum_{j=1}^d w_j (x_j - y_j)^2}$, used by a precision-farming platform to compare soil-sample feature vectors.

- (a) Prove that d_w is a genuine metric on $\mathbb{R}^d$. (*Hint:* let $x'_j = \sqrt{w_j} x_j$ for each j ; show $d_w(x, y) = d_2(x', y')$, the ordinary Euclidean distance between the rescaled vectors, and invoke the already-established triangle inequality for d_2 .)
- (b) Consider a query point $Q = (0, 0)$ (representing zero deviation from target pH and target Nitrogen level) and two candidate soil samples $A = (1, 4)$ and $B = (3, 1)$ (deviations in the same two units). Compute the unweighted Euclidean distances $d_2(Q, A)$ and $d_2(Q, B)$ and state which sample is the nearer neighbour.
- (c) Now apply weights $w = (5, 0.5)$ (emphasizing the pH-deviation coordinate five-fold relative to Nitrogen). Compute $d_w(Q, A)$ and $d_w(Q, B)$, and show explicitly that the nearest-neighbour ranking **flips** relative to part (b).
- (d) State, in one or two precise sentences, the general lesson this example establishes about applying k -NN-based imputation or classification to heterogeneous agronomic features without an explicit, justified feature-weighting (or scaling) decision.

[8 marks]

Q2. Soil moisture readings (%) from $n = 9$ field sensors on one plot are:

22, 24, 25, 26, 27, 28, 29, 31, 85.

Compute Q_1 , the median, Q_3 , and IQR via the median-of-halves convention, and hence the Tukey fences. State which observation(s) are flagged, and suggest (in one sentence, no marks lost either way) the most likely physical cause given the magnitude of the flagged reading.

[4 marks]

Q3. An irrigation-sensor network suffers a fault: a fraction $\varepsilon \in (0, 1)$ of readings are logged as an identical, coordinated placeholder value M (a maximum-scale fault code), while the remaining readings are confined to a bounded range negligible relative to M as $M \rightarrow \infty$.

- (a) Derive $\mu(M)$, $\sigma(M)$, and show the ordinary z -score of a faulty reading converges to $\sqrt{(1 - \varepsilon)/\varepsilon}$ as $M \rightarrow \infty$.

- (b) Solve for the critical fraction ε^* at which this limit equals 3, and state the operational conclusion for a farm-monitoring engineer: what fraction of a batch being corrupted identically renders the classical 3σ rule permanently unable to flag it, however extreme M is?
- (c) Prove that the modified z -score based on median/MAD instead diverges to $+\infty$ as $M \rightarrow \infty$ for any fixed $\varepsilon < 0.5$. State precisely, and distinguish clearly, the two different senses of “breakdown point” at play here (single contaminating point with $n \rightarrow \infty$ versus fixed contamination fraction ε with $M \rightarrow \infty$), and show they yield numerically different thresholds for this network.

[9 marks]

Q4. Two farm records are described by (Field Size in hectares, Fertilizer Expenditure in paise): $P = (2.1, 45000)$, $Q = (3.8, 62000)$.

- (a) Compute the raw Euclidean distance and identify the dominating coordinate.
- (b) Given regional statistics $\mu_{\text{size}} = 3.0, \sigma_{\text{size}} = 0.6$ and $\mu_{\text{exp}} = 50000, \sigma_{\text{exp}} = 9000$, standardize both coordinates and recompute the Euclidean distance.
- (c) A “similar farms” recommender for input-subsidy targeting is run on raw features. Explain, quoting your two distances, why recording expenditure in paise rather than rupees would silently make field size irrelevant to the recommendation.

[6 marks]

Q5. Two farms’ logged crop-issue tags over {PestDamage, Drought, Waterlog, NutrientDeficiency, FungalInfection} are $x = (1, 1, 0, 0, 1)$ and $y = (1, 0, 0, 1, 1)$.

- (a) Compute the Jaccard similarity and cosine similarity of x and y .
- (b) A third farm’s tag vector $z = (0, 0, 0, 0, 0)$ reflects a field-visit report that was never filed, not a genuine claim of zero issues. State $\cos(z, y)$ and $\cos(z, z)$ under the course convention, and explain why silently treating z as genuine “no issues” would corrupt an issue-similarity-based advisory clustering tool.

[4 marks]

Q6. Let R be the missingness indicator for a column $Y = (Y_{\text{obs}}, Y_{\text{mis}})$, with mechanism $P(R \mid Y_{\text{obs}}, Y_{\text{mis}}; \phi)$.

- (a) State Rubin’s formal definitions of MCAR, MAR, and MNAR.
- (b) Prove the ignorability result under MAR (with distinct parameters), showing the observed-data likelihood factors cleanly.
- (c) A state agricultural survey finds that farmers with the *worst* crop yields are least likely to self-report yield figures (for tax and subsidy-eligibility reasons), so $P(R = 0 \mid \text{Yield})$ increases as true Yield decreases. Classify this mechanism formally, and use your proof in (b)–(c) to explain concretely why no imputation using only observed survey fields can, even in principle, remove the resulting upward bias in average reported yield.
- (d) Suppose the survey agency also has an always-observed satellite-derived vegetation index (NDVI) correlated with true yield. Explain, in general terms, how conditioning on this variable could move the analysis toward a recoverable MAR-type situation relative to the enlarged observed set, and why this shows the classification is relative to the observed variable set, not intrinsic to Yield itself.

[8 marks]

- Q7.** An analyst applies iterative predictive imputation to five farm records over Field Size (ha), Rainfall (mm), and Yield (t/ha):

Row	FieldSize	Rainfall	Yield
1	2.5	850	4.2
2	3.1	780	?
3	1.8	900	3.6
4	?	700	5.0
5	4.0	820	4.6

Missing cells are initialized with the column mean over observed values. The cyclic order is **Yield** $\rightarrow$ **FieldSize**, using the fitted relations

$$\widehat{\text{Yield}} = 0.002 \cdot \text{Rainfall} + 0.5 \cdot \text{FieldSize} + 2, \quad \widehat{\text{FieldSize}} = 0.3 \cdot \text{Yield} - 0.001 \cdot \text{Rainfall} + 2.$$

- State the two initial fill values.
- Perform one full iteration (Row 2's Yield, then Row 4's FieldSize), showing all substituted numbers.
- Explain, referencing the specific predictor columns involved, why a second iteration leaves both values unchanged here, and state a general condition on predictor overlap under which it would not.

[6 marks]

- Q8.** A farmer-identity-resolution system linking the State Agricultural Database and Bank KYC records computes pairwise similarities for three farmer records A, B, C : $s(A, B) = 0.77$, $s(B, C) = 0.75$, $s(A, C) = 0.16$, with lower threshold 0.4 and upper threshold 0.68.

- Classify each pair.
- State the connected-components clustering of $\{A, B, C\}$ under “match” edges, and explain the transitivity problem.
- With $\text{Obj}(\mathcal{C}) = \sum_{i,j \text{ same cluster}} (s(i, j) - \theta)$, $\theta = 0.5$, compute Obj for $\{A, B, C\}$ versus $\{A, B\}$, $\{C\}$, and conclude which is objectively preferable.
- For a chain of n farmer records with adjacent similarity s_0 (just above upper threshold) and non-adjacent similarity floor f (below lower threshold), derive Obj exactly for (i) the single connected-component clustering and (ii) the clustering retaining only adjacent pairs, and prove the gap is $\Theta(n^2)$. State the operational lesson for a state agency merging farmer identities across many independently digitized district land records.

[10 marks]

- Q9.** Outstanding farm-loan amounts (in Rs.) for nine accounts are strongly right-skewed: 6, 10, 15, 22, 32, 58, 110, 280, 1100. Compute $\log_{10}$ of each to two decimal places, and state the raw ratio max / min against the transformed range, quantifying the compression achieved.

[4 marks]

- Q10.** An agri-fintech team fits a scaler for *Farm Age (years since establishment)* and a one-hot encoder for *Soil Type* on the entire dataset (training and test rows combined) before splitting, to predict crop-failure risk for loan underwriting.

- (a) With $n_{\text{tr}}, n_{\text{te}}$ the train/test sizes and $\mu_{\text{tr}}, \mu_{\text{te}}$ the true population Farm-Age means restricted to each subpopulation (which may differ if the test set is drawn from a newly onboarded district), derive the exact bias of the leaked full-data mean relative to the correct train-only mean, and state the condition under which it vanishes.
- (b) Explain formally why this bias implies any downstream test-set risk-prediction metric is optimistically biased.
- (c) A Soil Type value (“Laterite”) appears only among test-set farms from a newly onboarded region. State precisely how a leak-free encoder fitted on training soil types only must represent this row, and explain why fitting the encoder on the union of all soil types seen across the whole dataset is still leakage.

[8 marks]

Q11. Two candidate duplicate farmer records are compared on four fields: name similarity = 0.91 (Jaro–Winkler), Aadhaar-last-4 agreement = 1, phone agreement = NaN (missing on one record), village agreement = 0. Weights: name 0.4, Aadhaar 0.3, phone 0.2, village 0.1.

- (a) Compute the weighted-mean score using only available fields, weights renormalized.
- (b) Classify against thresholds lower = 0.5, upper = 0.75.
- (c) Explain why substituting 0 for the missing phone field instead of excluding and renormalizing would bias the decision, and in which direction.

[6 marks]

Q12. Crop-plant counts per acre for $n = 7$ sampled plots: 48, 50, 49, 51, 48, 50, 310.

- (a) Compute the mean, standard deviation, and ordinary z -score of the value 310.
- (b) Compute the median, MAD, and modified z -score of 310.
- (c) Using $|z| \geq 3$ and $|z_{\text{mod}}| > 3.5$, state which method(s) flag 310, and connect the disagreement explicitly to your general result in Q3.

[4 marks]

Q13. Let X be a numerical feature with mean μ_X , standard deviation $\sigma_X > 0$, ordinary z -score $z_X(x) = (x - \mu_X)/\sigma_X$, and Tukey fences $[Q_1 - 1.5 IQR, Q_3 + 1.5 IQR]$. Consider the affinely transformed variable $X' = aX + b$ for constants $a \neq 0, b$.

- (a) Prove that for $a > 0$, $z_{X'}(ax + b) = z_X(x)$ for every x (i.e. the z -score of the transformed observation equals the z -score of the original observation).
- (b) Prove that for $a < 0$, $z_{X'}(ax + b) = -z_X(x)$, and explain why the flagging rule $|z| \geq 3$ is nonetheless unaffected by the sign of a .
- (c) Prove that for $a > 0$, every quantile transforms as $Q_p(X') = a Q_p(X) + b$, and hence that $IQR(X') = a \cdot IQR(X)$ and the Tukey fences transform as $\text{Fence}(X') = a \cdot \text{Fence}(X) + b$. Conclude that a reading flagged as an outlier on the raw scale is flagged if and only if its image is flagged on the X' scale, for any $a > 0, b$.
- (d) A farm-sensor vendor recalibrates a device by applying $X' = -X + 100$ (e.g. converting a “distance-from-full-scale” reading into a “percentage-remaining” reading). Using parts (a)–(b), state precisely what happens to the ordinary z -score of a given physical reading under this recalibration, and explain why an outlier-detection pipeline that hard-codes the *sign* of a z -score threshold (rather than its absolute value) would silently break after this recalibration.

[8 marks]

Q14. Two blocking passes generate candidate duplicate farmer pairs among $n = 5000$ standardized records linking the State Agricultural Database and Bank KYC systems. Pass 1 (same district and birth year) yields blocks of sizes 48, 65, 32. Pass 2 (same last-four registered-phone digits) yields blocks of sizes 9, 12, 7, 16. The two candidate sets overlap in 45 pairs.

- Compute the candidate pairs from each pass and the size of their union.
- Compute the reduction ratio RR for the union relative to $\binom{n}{2}$.
- Ground truth contains 320 true duplicate pairs; Pass 1 alone captures 270, Pass 2 alone captures 170, both jointly capture 130. Verify via inclusion–exclusion the pair completeness of the union and state its numerical value.
- Prove $1 - (1 - p_1)(1 - p_2) \geq \max(p_1, p_2)$ for independent capture probabilities $p_1, p_2 \in [0, 1]$, and relate it to your answer in (c).

[6 marks]

Q15. A state agricultural data officer gives you eight farm records:

ID	FieldSize (ha)	Rainfall (mm)	Yield (t/ha)	Farmer Name (std.)
1	2.5	850	4.2	ramesh patil
2	2.2	800	4.0	ramesh patil
3	4.1	870	?	suresh gowda
4	5.0	900	5.1	manoj yadav
5	3.0	1400	1.8	anita devi
6	4.8	890	?	d. reddy
7	4.0	865	4.8	suresh gowda
8	2.4	845	4.1	kavita rao

Among the six rows with observed Yield, mean FieldSize is 3.183; among the two with missing Yield, mean FieldSize is 4.45. From the other seven rows (excluding Row 5), $\mu = (\mu_R, \mu_Y) = (858.6, 4.24)$ for (Rainfall, Yield) and $\Sigma = \begin{bmatrix} 520 & -8 \\ -8 & 0.16 \end{bmatrix}$. Rows 3 and 7 are suspected duplicate farmer records, with comparison vector: name similarity 0.95, Aadhaar NaN, village agreement 1, DOB agreement 0, using the weights/thresholds of Q11.

- State, with the standard caveat about MNAR being unprovable from observed data alone, what the field-size-conditioned evidence does and does not establish about the missingness mechanism of Yield.
- Compute, by hand, the Mahalanobis distance of Row 5's (Rainfall, Yield) = (1400, 1.8) from μ , explicitly computing Σ^{-1} , and comment on whether Row 5 is a multivariate outlier despite each coordinate looking merely “high” or “low” in isolation rather than impossible.
- Decide, via the Q11 machinery, whether Rows 3 and 7 should be merged, and construct the merged record's Yield field using the course's numeric golden-record rule.
- Argue formally whether deduplication should precede or follow imputation here, prove your ordering strictly dominates the reverse for at least one field, and state the audit-trail invariant that must survive regardless of ordering.

[9 marks]